KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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Software engineer with production experience in Laravel/Vue.js and hands-on experience building LLM, retrieval, and multimodal AI systems in Python.

EXPERIENCE

Kahap Inc | *Software Developer (Part-time)* Aug 2022 - Aug 2026

- Built and maintained a production feedback platform, including submission workflows, automated notifications, and admin tools.
- Developed Laravel/Vue.js features and debugged application and database issues in production.
- Handled deployments and server operations, including server upgrades, Laravel queues/workers, SSL renewal, and database troubleshooting.

Pac Com Solution | *Full-Stack Developer Intern* Jun 2021 - Jan 2022

- Developed and maintained Laravel/Vue.js features for production POS applications.

PROJECTS

Semantic Text-to-SQL | *Python · XiYanSQL 7B · LangGraph · FastAPI · SQLite*

- Built a local Text-to-SQL service with XiYanSQL 7B, schema grounding, SQL validation, read-only execution, and bounded repair.
- Evaluated XiYanSQL 7B on 100 BIRD questions, matching the expected execution result on 58, then used failure analysis to compare schema-linking approaches.

Local RAG Search | *Python · BM25 · Qwen3-Embedding-0.6B · RRF · FastAPI*

- Built a local hybrid search service combining BM25, Qwen3 embeddings, and RRF behind a FastAPI API.
- On 64 EnterpriseRAG-Bench queries over 5,189 documents, RRF outperformed BM25 on Recall@10 (0.763 vs 0.734) and MRR@20 (0.742 vs 0.679).

Hermes Video Tutor | *Python · Hermes · Qwen3.5 4B · Tool Calling · Multimodal*

- Built a video-QA agent that answers from transcripts when possible and inspects frames or clips only when visual evidence is needed.
- Tested on 12 questions, adding visual tools improved overall pass rate from 16.7% to 50.0%, with 66.7% on visual questions.

SKILLS

AI / LLM: LLM Agents, RAG, Tool Calling, Retrieval, Evaluation, LangGraph

Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js, React

Data / Infra: AWS, MySQL, Redis, SQLite, Linux, Git, pytest

Other: Java, C/C++ | Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | *M.S., Information Systems & Applications* Sep 2024 - Jun 2026

GPA 4.09/4.3 | Taiwan | Thesis: *Interpreting Implicit Intent through Workplace Hierarchies and Context*

Chung Hua University | *B.S., Computer Science & Information Engineering* Sep 2022 - Jun 2024

GPA 3.9/4.0 | Taiwan